Differentially Expressed Genes Associated with Glaucoma

By

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Abstract

Glaucoma is the leading cause of irreversible blindness worldwide, with a multifactorial aetiology that includes genetic components not yet fully understood. This study aimed to explore differentially expressed genes (DEGs) in glaucoma compared to a group with no eye pathology. Two datasets containing transcriptomics data from the trabecular meshwork and astrocytes of the optic nerve head were selected from the public database Gene Expression Omnibus (GEO) and analysed using the in-built limma package of R. The Benjamini and Hochberg false discovery rate was applied to determine significance, with a p-value ≤ 0.05 considered significant. Genes with a log two-fold change ≥ 1 were recognised as upregulated, while those ≤ -1 were considered downregulated in the glaucoma cohort. The ten most upregulated and downregulated DEGs were selected for further analysis. A protein-to-protein interaction network with standard settings was generated using the Search Tool for the Retrieval of Interacting Genes/Proteins (STRING). Functional annotation for Gene Ontology and pathway enrichment analysis was performed using the Database for Annotation, Visualization and Integrated Discovery (DAVID) with default parameters. Upregulated DEGs in the trabecular meshwork were enriched in haemoglobin-related processes, while downregulated were in structural and extracellular matrix-associated mechanisms, and in immunologic responses. The enrichment of DEGs in astrocytes of the optic nerve head was associated with metabolic processes. Although, any of the DEGs may be further explored as therapeutic target, AKR1C3, CALML3, and ENO2 should be prioritised. Additionally, ENO2 also showed promise for diagnostic applications.

Keywords

Glaucoma, differentially expressed genes, functional enrichment analysis.

Abbreviations

RGCs, retinal ganglion cells; IOP, intraocular pressure; POAG, primary open-angle glaucoma; TM, trabecular meshwork; DEGs, differentially expressed genes; ONH, optic nerve head; C, Caucasian; AA, African American; FDR, false discovery rate; STRING, Search Tool for the Retrieval of Interacting Genes/Proteins; PPI, protein-to-protein interaction; DAVID, Database for Annotation, Visualization and Integrated Discovery; GO, gene ontology; KEGG, Kyoto Encyclopedia of Genes and Genomes; BP, biological processes; MF, molecular functions; CC, cellular compartments; ECM, extracellular matrix; Hb, haemoglobin; ROS, reactive oxidative species; BMP2, bone morphogenetic protein 2; TGF-β, transforming growth factor-beta; ALDH, aldehyde dehydrogenase; MMP-9, matrix metalloproteinase 9; LOXL1, lysyl oxidase like 1.

Introduction

Glaucoma, a group of neurodegenerative diseases involving the eye, is the leading cause of irreversible visual impairment worldwide, with 3.6 million patients aged over 50 affected in 2020 (Steinmetz, 2021). Tham *et al*. (2014) estimated 80 million people suffering from glaucoma in the same year, increasing over 111 million by 2040.

This devastating blinding condition is the consequence of the continuous deterioration of the retinal ganglion cells (RGCs), and ultimately, of the optic nerve (Yang and Sun, 2023), which impairs the adequate transmission of the visual information from the eye to the brain. The overall glaucoma pathogenesis is to a certain extent known. Nonetheless, there are multiple factors that may be involved in its development, and these are poorly understood. For example, it is known that the major cause of RGCs decline is due to the abnormal and persistent increase in the intraocular pressure (IOP) that occurs when there is an excessive presence of aqueous humour due to its exacerbated production or unsatisfactory drainage, mainly through the trabecular meshwork (TM). However, the mechanisms that trigger these outcomes are still unrevealed. Other well known risk factors are advanced age, presence of comorbidities such as diabetes, family history of glaucoma or other eye diseases, an increased cup-to-disc ratio and asymmetry, and there is a certain increased prevalence depending on ethnic origin (Hollands *et al*., 2013; Mcmonnies, 2017). Recent studies highlighted the involvement of other features related to its pathogenesis such as vasoconstriction factors (Lommatzsch *et al*., 2022; Zhou and Sabel, 2023), autoimmune reactions (Wakefield and Wildner, 2020) or intracranial pressure due to cerebrospinal fluid alterations (Fleischman and Allingham, 2013). Despite these elements may also play a role in the development and progression of glaucoma, the only modifiable risk factor is the IOP, which remains the sole focus of therapeutic management for this disorder (Wagner *et al*., 2022), mainly through the use of eye drops and surgical laser treatments. However, these treatments often result in adverse effects or long-term ineffectiveness (Lusthaus and Goldberg, 2019). In addition, an early detection of glaucoma remains one of the major challenges due to its prolonged asymptomatic phase, becoming only evident after the disease has progressed significantly, complicating its prompt treatment and favouring the irreversible consequences in the optic nerve. It is clear that there is a need to develop new diagnostic and therapeutic approaches to tackle the current challenges and improve the quality of life for patients affected by glaucoma.

One of the aspects that has gained more attention in the last decade is the genetic implication around the development and progression of the disease. The involvement of several genes has

already been described in the literature, although the exact mechanisms influencing their role in glaucoma are not fully understood yet (Wang *et al*., 2022). Sinnasamy and colleagues (2022) have summarised some of the most relevant genes and the Human Genome Organisation, HUGO (nd) currently has in their database 67 genes associated with glaucoma. In addition, another promising route to gain more understanding about what molecular mechanisms are implicated in the glaucoma pathogenesis and progression, is the identification and study of differentially expressed genes (DEGs) involved in the disease. Moreover, the examination of these DEGs may provide insight into biological processes and molecular patterns with protective capabilities, which could be further explored as potential molecular glaucoma therapies.

The study presented within this article was aimed at the identification of the most significant DEGs in the two anatomical structures most affected by glaucoma, the trabecular meshwork and the optic nerve. The goal was to explore their potential involvement in the pathogenesis and progression of the disease and identify future novel molecular markers that might have diagnostic and therapeutic implications.

Materials and Methods

This study did not include any animal or human experimentation by the authors and was performed under the approval of the Ulster University School of Biomedical Sciences Ethics Filter Committee.

Array data collection

The web-based repository Gene Expression Omnibus (GEO; https://www.ncbi.nlm.nih.gov/gds), hosted by the National Center for Biotechnology Information (NCBI) (Edgar *et al*., 2002), was used to search for suitable datasets. Two separate searches were performed using the keywords “primary open angle glaucoma” and “glaucoma”, filtering on both occasions by organism “*Homo sapiens”* and study type “expression profiling by array”. Two gene expression profile datasets, GSE27276 platform GLP2507 Sentrix Human-6 Expression BeadChip (Liu *et al.,* 2013) and GSE9944 platform GLP8300 [HG_U95AV2] Affimetrix Human Genome U95 Version 2 Array (Lukas *et al.,* 2008), were included for further analysis. Very low sample size datasets, studies that included drug treatments, and datasets formed by samples of origin other than trabecular meshwork (TM) or optic nerve head (ONH), were excluded.

GSE27276 was composed of 17 TM samples from patients diagnosed with primary open-angle glaucoma (POAG) and 19 TM samples from control donors without eye pathology. The TM samples from POAG patients were obtained during surgery, while control specimens were

retrieved from donated eyes following the same procedure as applied during the trabeculectomy surgery. Postmortem delay was 0 hours for the POAG specimens and between 2.25 – 6.24 hours for the control tissue (Liu *et al.,* 2013). No other demographic data was available from this dataset.

Dataset GSE9944 comprised 13 ONH samples from nine patients diagnosed with glaucoma, six Caucasian and three African American, and six with no known eye disease from six Caucasian individuals. Ages ranged from 53-83 and 42-72, respectively in these two cohorts. ONH specimens were taken from both eyes in four out of nine individuals with glaucoma.

A description of the available demographic data for both datasets is included in **Table A1** and **Table A2** in the Appendix.

Identification of Differentially Expressed Genes (DEGs)

The gene expression of the datasets was analysed separately using the NCBI GEO2R tool (https://www.ncbi.nlm.nih.gov/geo/geo2r/) and the limma package v3.54.00 (Smyth, 2005). Benjamini and Hochberg (1995) and false discovery rate (FDR) (Benjamini and Hochberg, 2000) were used to determine the p-value, which was adjusted to ≤ 0.05. Log2 fold change |log2FC| threshold was adjusted to ≥ 1 for upregulated and ≤ -1 for downregulated genes. A list containing the DEGs was generated and downloaded. The top ten upregulated and downregulated DEGs were selected for further analysis. Volcano plots and mean difference plots were also generated for each dataset using the same parameters as in the GEO2R analysis. A heatmap with the gene expression of the top ten upregulated and downregulated DEGs was created per array using the interactive web server Heatmapper (www.heatmapper.ca/expression/) (Babicki *et al.,* 2016). Two Venn diagrams were generated using the online server Biotools (https://www.biotools.fr/misc/venny) (Saurin, 2024) to assess separately the coincidence of upregulated and downregulated DEGs between the two datasets.

Protein-to-Protein Interaction (PPI) and Functional Annotation Analysis

Two protein networks per dataset, corresponding to the ten most upregulated and downregulated DEGs, were created using the Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) v12.0 (https://www.string-db.org/) (Szklarczyk *et al.,* 2019). Default settings, including a confidence interaction score of ≥ 0.4, were used to study this interaction.

The functional enrichment analysis of the Gene Ontology (GO) terms including biological processes (BP), molecular functions (MF), cellular components (CC), and the Kyoto

Encyclopedia of Genes and Genomes (KEGG) Pathway were performed using the Database for Annotation, Visualization and Integrated Discovery (DAVID) webserver v2024q1 (https://davidbioinformatics.nih.gov/summary.jsp) (Dennis *et al.,* 2003; Sherman *et al.,* 2022), using the Functional Annotation Clustering tool with a medium classification stringency and default values; kappa similarity overlap set to 3, and threshold of 0.5; a classification membership of initial and final group of 3, and multiple threshold linkage set to 0.5. The modified Fisher exact p-value (EASE score) was set to ≤ 0.05 and the Benjamini and Hochberg method (1995) and FDR (Benjamini and Hochberg, 2000) were selected. Finally, results obtained from DAVID analysis were plotted using bubble graph generated with SRplot (https://www.bioinformatics.com.cn/en?keywords=bubble) (Tang *et al*., 2023).

Results

Data Collection and Selection of DEGs

The two separated GEO queries using the keywords “primary open angle glaucoma” and “glaucoma” showed 14 and 35 results, respectively, of which some results overlapped. GSE27276 and GSE9944 were selected, and the composition of each dataset can be seen in the Appendix (**Table A1** and **Table A2**, respectively). In the dataset GSE27276 197 DEGs were identified with adjusted p-value ≤ 0.05 and |log2FC| cutoff of ≥ 1 for upregulated and ≤ -1 for downregulated genes; 64 genes were upregulated and 133 were downregulated. The ten most upregulated genes in POAG compared to controls were HBA2, HBA1, HBB, HBD, MGP, FCGBP, C2orf40, HLA-DPA1, CYTL1 and FMOD, while KRT19, CALML3, KRT15, KRT13, C19orf33, PAX6, SLPI, LCN2, S100A9, S100A8 were the ten most downregulated DEGs. **Table 1** shows the data from the top ten upregulated and downregulated DEGs of this dataset.

Dataset GSE9944 had 15 significant DEGs; twelve upregulated and three downregulated. The ten most upregulated DEGs in glaucoma compared to the control group were AKR1C3, SEPP1, ADH1B, DCLK1, LY96, KCTD12, MID1, RBP1, ADH1A and CEBPD, while the only three downregulated DEGs were ELN, ENO2 and BMP1. **Table 2** shows this distribution and **Figure 1** and **Figure 2** represent the volcano and mean difference plots for each dataset. **Figure 3** demonstrates the expression level of the ten most significant DEGs both upregulated and downregulated for datasets GSE27276 (**Figure 3. A**) and GSE9944 (**Figure 3. B)**. Venn diagrams **(Figure 4)** demonstrated no gene overlap between the two datasets for both upregulated and downregulated DEGs.

PPI Networks and DAVID Analysis

The network corresponding to the top ten upregulated DEGs from GSE27276 comprised a total of ten nodes and seven edges, formed by two clusters and four independent proteins: FCGBP, C2orf40 (ECRG4), CYTL1 and HLA-DPA1. The cluster with the highest number of interactions is composed of four upregulated proteins: HBA2, HBA1, HBB, and HBD with six edges connecting them. The second most significant cluster had only one edge linking two proteins; FMOD and MGP (**Figure 5. A**). The network generated with the top ten downregulated DEGs for GSE27276 was formed by ten nodes and eight edges with two clusters and three independent proteins: CALML3, C19orf33 and PAX6. The most significant cluster comprised four nodes and five edges interacting between S100A8, LCN2, SLPI, and S100A9, the latter also directly linked to LCN2. The second most relevant cluster consisted of three nodes and three edges linking KRT15, KRT19 and KRT13 (**Figure 5. B**).

The top ten upregulated DEGs from the dataset GSE9944 generated only one cluster composed of three nodes and two edges connecting ADH1B and RBP1 to ADH1A. The remaining seven nodes; CEBPD, MID1, DCLK1, LY96, KCTD12, AKR1C3, and SELENOP were independent (**Figure 6. A**). The only three significant downregulated DEGs for this array formed a cluster linking BMP1 and ELN while ENO2 remained independent (**Figure 6. B**).

Gene Ontology analysis by DAVID revealed that four of the most significant upregulated genes for the dataset GSE27276, HBA1, HBA2, HBB, HBD, were involved in four BP; carbon dioxide transport (GO: 0015670), oxygen transport (GO: 0015671), hydrogen peroxide catabolic process (GO: 0042744), cellular oxidant detoxification (GO: 0098869). In three CC terms; haptoglobin-haemoglobin complex (GO: 0031838), blood microparticle (GO: 0072562), haemoglobin complex (GO: 0005833), and in six MF terms; oxygen transporter activity (GO: 0005344), oxygen binding (GO: 0019825), haptoglobin binding (GO: 0031720), peroxidase activity (GO: 0004601), haem binding (GO: 0020037), organic acid binding (GO: 0043177). HBA1, HBA2 and HBB alone had involvement in two BP; response to hydrogen peroxide (GO: 0042542), and nitric oxide transport (GO: 0030185). In one CC term; endocytic vesicle lumen (GO: 0071682), and in two KEGG; malaria (hsa05144), and African trypanosomiasis (hsa05143). In addition, HBB and HBD participated in the MF term haemoglobin alpha binding (GO: 0031721). FCGBP, MGP, HBB, HBA1, HBA2 in the CC term extracellular exosome (GO: 0070062), FCGBP, HBB, HBA2, CYTL1, HBA1, FMOD in extracellular space (GO: 0005615) and FCGBP, MGP, FMOD in the extracellular matrix (GO: 0031012). Three of the ten most downregulated genes for this dataset, KRT19, KRT15, and KRT13, had participation in two BP; intermediate filament organization (GO: 0045109) and epithelial cell differentiation (GO: 0030855). In one CC term;

intermediate filament (GO: 0005882), one MF term; structural molecule activity (GO: 0005198), and one KEGG pathway; Staphylococcus aureus infection (hsa05150). LCN2, S100A9, and S100A8 were involved in one BP term; defence response to bacterium (GO: 0042742), and in one KEGG pathway; oestrogen signalling pathway (hsa04915). SLPI, LCN2, S100A9, S100A8 in the BP term; innate immune response (GO: 0045087), SLPI, S100A9, S100A8 in another BP term; response to lipopolysaccharide (GO: 0032496), KRT19, KRT15, KRT13, S100A9, S100A8 in one CC term; cytoskeleton (GO: 0005856), KRT19, SLPI, KRT15, KRT13, LCN2, CALML3, S100A9, S100A8 in another CC term; extracellular exosome (GO:0070062), and KRT19, KRT15, KRT13, CALML3 in the KEGG IL-17 signalling pathway (hsa04657).

Enrichment analysis of the ten most significant upregulated genes for the dataset GSE9944 resulted in two functional terms; one BP and one CC; retinol metabolic process (GO: 0042572) involving the genes ADH1B, ADH1A, RBP1, AKR1C3, and retinol dehydrogenase activity (GO: 0004745) with participation of ADH1A, ADH1B, AKR1C3. This analysis also revealed the involvement of ADH1A, ADH1B, and LY96 in one KEGG pathway; alcoholic liver disease (hsa04936). No downregulated DEGs were enriched in this dataset. **Figure 7** describes the DAVID analysis for both datasets and functional annotation raw data from the experiment is included in **Table A3** in the Appendix.

Table 1. GSE27276 top ten upregulated and downregulated genes.

GEO ID	Adj. p-value	LogFC	Gene Symbol	Gene Title
Upregulated				
GI_14043068-S	0.00	5.433	HBA2	Haemoglobin subunit alpha 2
GI_14456711-S	0.00	5.053	HBA1	Haemoglobin subunit alpha 1
GI_28302128-S	0.00	4.976	HBB	Haemoglobin subunit beta
GI_6633803-S	0.00	3.542	HBD	Haemoglobin subunit delta
GI_4505178-S	0.00	2.387	MGP	Matrix Gla protein
GI_4503680-S	0.00	2.149	FCGBP	Fc fragment of IgG binding protein
GI_14165275-S	0.00	1.976	C2orf40 (ECRG4)	Chromosome 2 open reading frame 40
GI_24797073-S	0.00	1.843	HLA-DPA1	Major histocompatibility complex, class II, DP alpha 1
GI_8922107-S	0.00	1.694	CYTL1	Cytokine like 1
GI_5016093-S	0.00	1.642	FMOD	Fibromodulin
Downregulated				
GI_40217850-S	0.00	-4.3383	KRT19	Keratin 19
GI_36031099-S	0.00	-3.7412	CALML3	Calmodulin like 3
GI_24430189-S	0.00	-3.3909	KRT15	Keratin 15
GI_24234693-A	0.00	-3.3699	KRT13	Keratin 13
GI_27485203-S	0.00	-2.9547	C19orf33	Chromosome 19 open reading frame 33
GI_4505614-A	0.00	-2.922	PAX6	Paired box 6
GI_15834622-S	0.00	-2.8998	SLPI	Secretory leukocyte peptidase inhibitor
GI_38455401-S	0.00	-2.8835	LCN2	Lipocalin 2
GI_9845520-S	0.00	-2.7844	S100A9	S100 calcium binding protein A9
GI_21614543-S	0.01	-2.6599	S100A8	S100 calcium binding protein A8

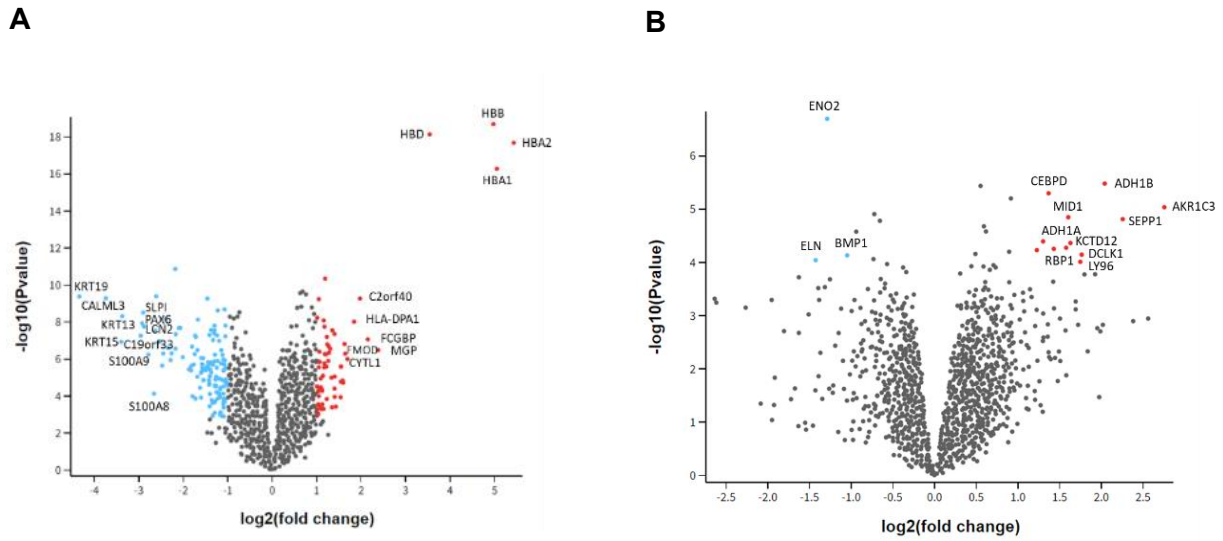
Adj. p-value = adjusted p-value; LogFC = log fold change

Table 2. GSE9944 top ten upregulated and downregulated genes.

GEO ID	Adj. p-value	LogFC	Gene Symbol	Gene Title
Upregulated				
37399_at	1.93E-02	2.755	AKR1C3	Aldo-keto reductase family 1, member C3
34363_at	2.08E-02	2.257	SEPP1	Selenoprotein P, plasma, 1
35730_at	1.54E-02	2.041	ADH1B	Alcohol dehydrogenase 1B (class I), beta polypeptide
38957_at	4.04E-02	1.765	DCLK1	Doublecortin like kinase 1
33956_at	4.72E-02	1.748	LY96	Lymphocyte antigen 96
38972_at	3.40E-02	1.630	KCTD12	Potassium channel tetramerization domain containing 12
34296_at	2.08E-02	1.604	MID1	Midline 1
38634_at	3.89E-02	1.577	RBP1	Retinol binding protein 1
34637_f_at	3.89E-02	1.430	ADH1A	Alcohol dehydrogenase 1A (class I), alpha polypeptide
1052_s_at	1.58E-02	1.368	CEBPD	CCAAT/enhancer binding protein delta
Downregulated				
31621_s_at	4.58E-02	-1.425	ELN	Elastin
40193_at	2.52E-03	-1.287	ENO2	Enolase 2
39407_at	4.04E-02	-1.047	BMP1	Bone morphogenetic protein 1

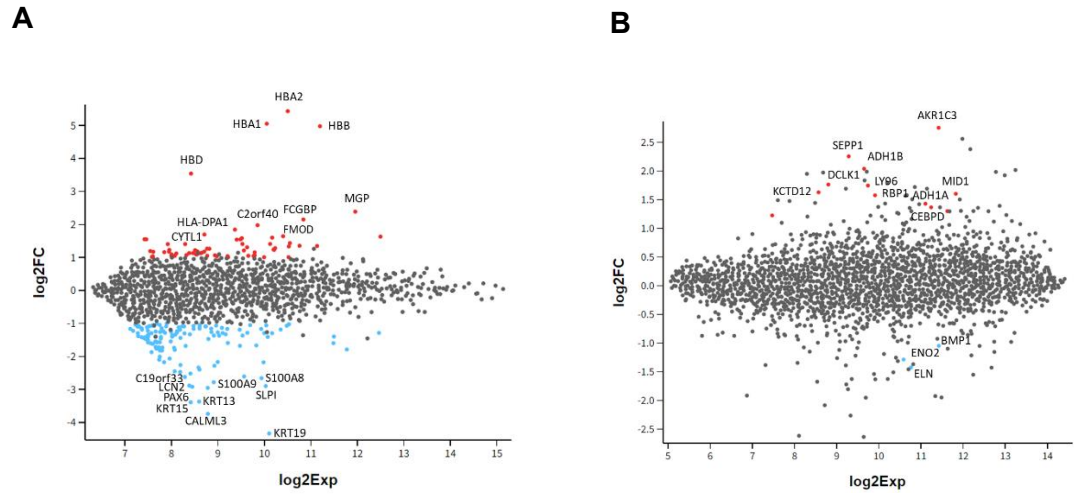
Adj. p-value = adjusted p-value; LogFC = log fold change

Figure 1. Volcano Plots. **A.** GSE27276 dataset. **B**. GSE9944 dataset.



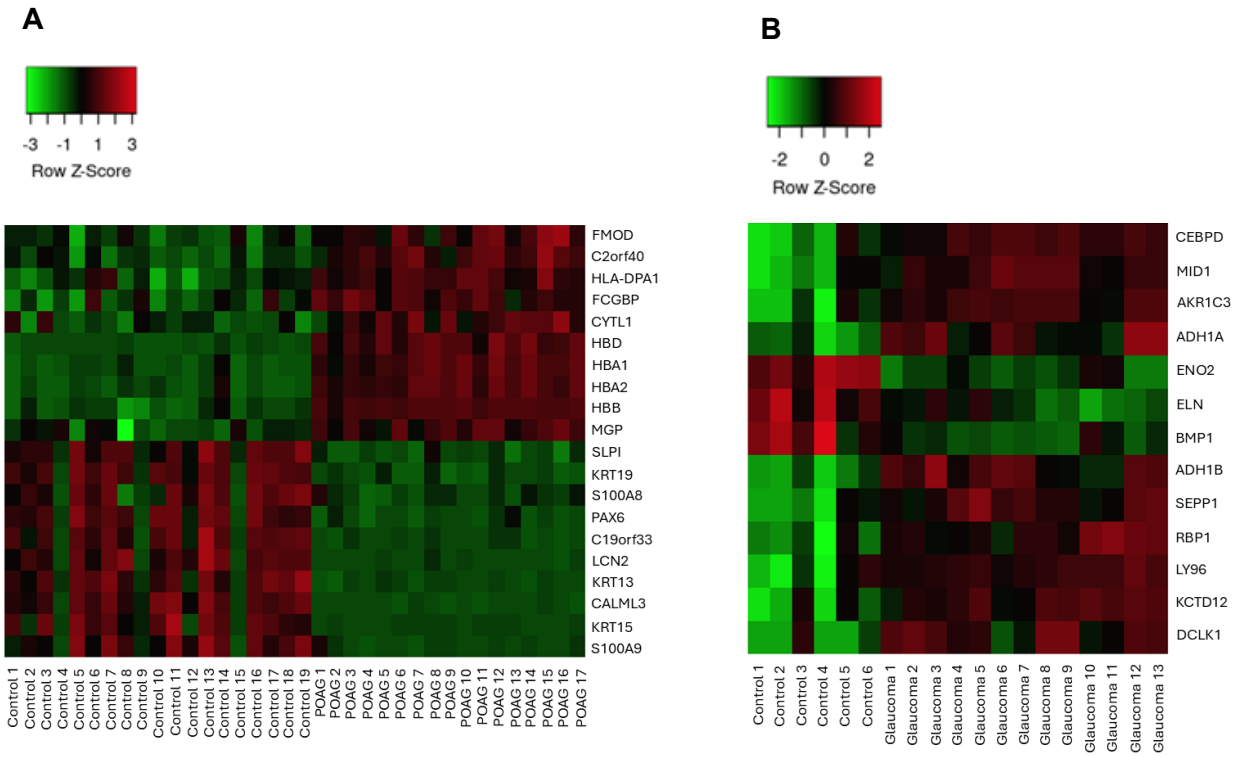
The plots compare the DEGs between glaucoma patients and control group. X-axis represents the |log2FC| and Y-axis the -log10 of a p-value. The p-value was adjusted to ≤ 0.05 and the |log2FC| to ≥ 1 for upregulated and ≤ -1 for downregulated DEGs. The top ten upregulated and downregulated DEGs are identified for both datasets. Significant upregulated genes are in red and downregulated genes are in blue. Black dots show unsignificant genes

Figure 2. Mean difference plots. **A.** GSE27276 dataset. **B**. GSE9944 dataset.



The plots compare the DEGs between glaucoma patients and control group. X-axis represents the expression levels of genes |log2Exp| and Y-axis the |log2FC|. The p-value was adjusted to ≤ 0.05 and the |log2FC| to ≥ 1 for upregulated and ≤ -1 for downregulated DEGs. The top ten upregulated and downregulated DEGs are identified for both datasets. Significant upregulated genes are in red and downregulated genes are in blue. Black dots show unsignificant genes.

Figure 3. Heatmaps for the ten most upregulated and the ten most downregulated genes**. A.** GSE27276 dataset. **B**. GSE9944 dataset.



The expression levels of upregulated DEGs in patients with POAG/glaucoma are represented in red, while downregulated DEGs are shown in green.

Figure 4. Venn diagrams. **A.** Upregulated DEGs. **B.** Downregulated DEGs.

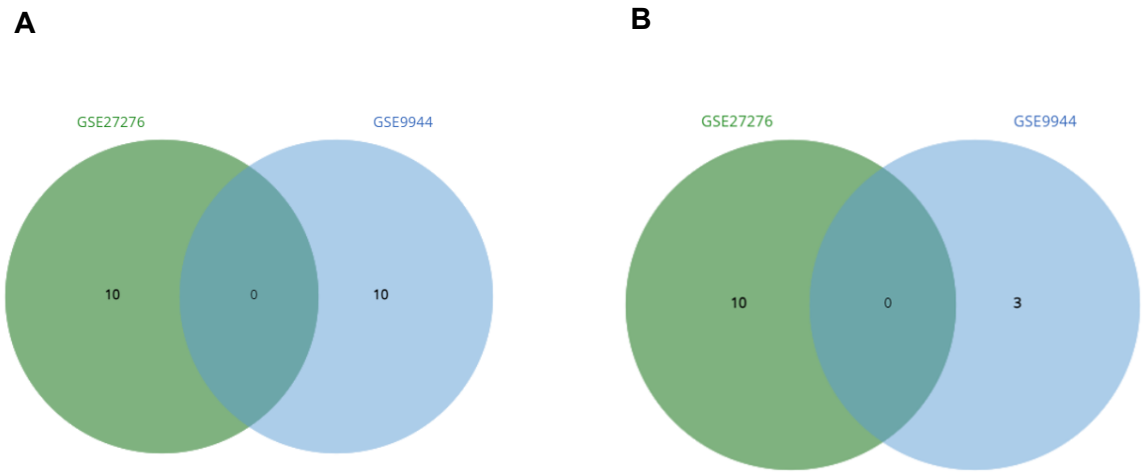
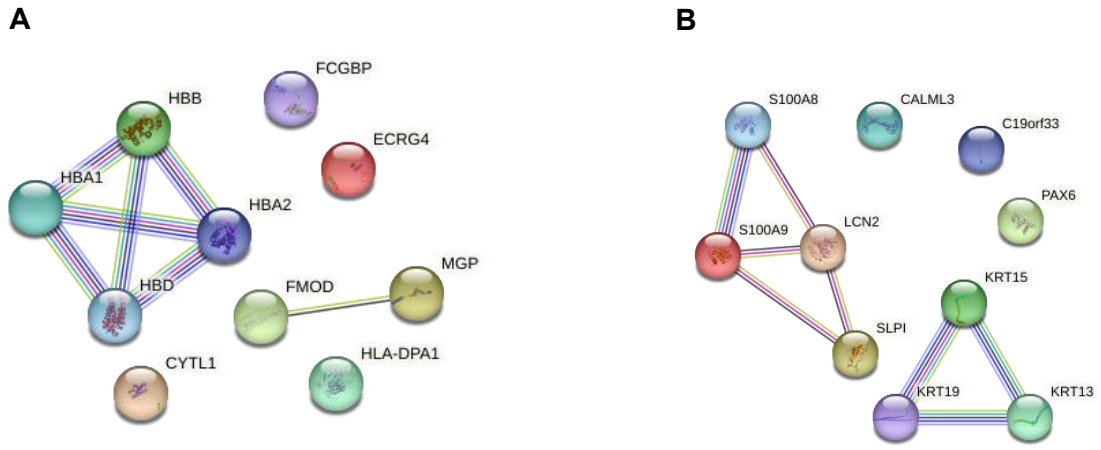
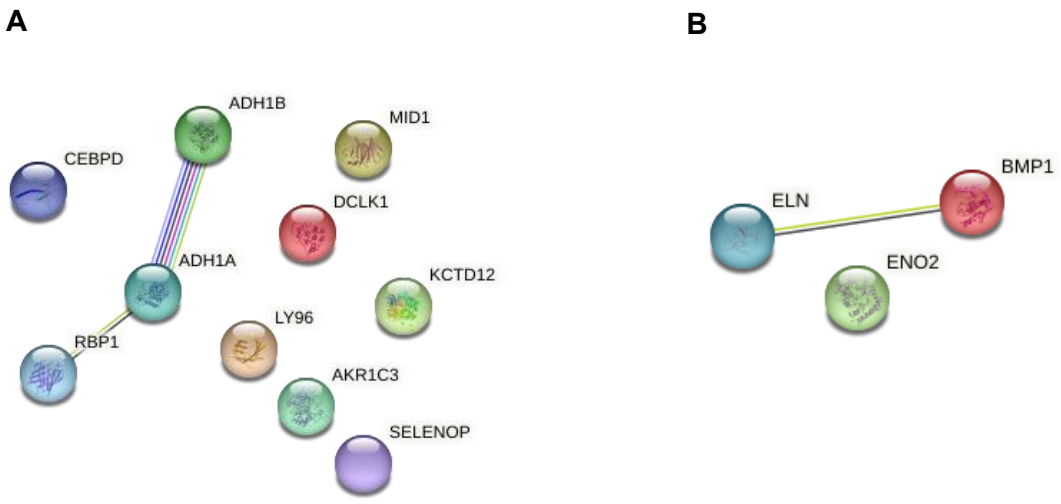


Figure 5. Protein-to-protein interaction networks for GSE27276 dataset. **A.** Top ten upregulated DEGs. **B**. Top ten downregulated DEGs.



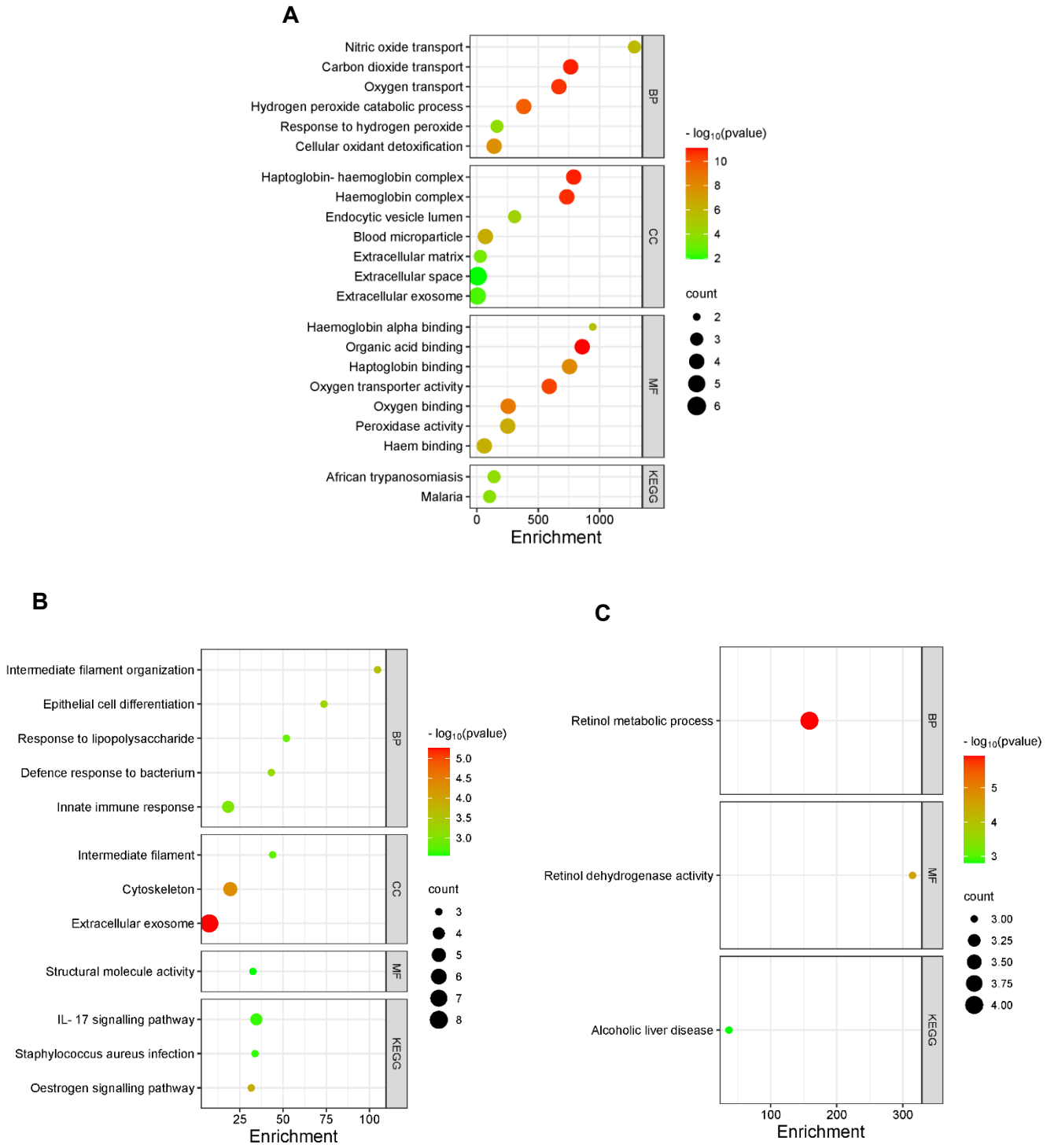
Each coloured circle represents a node. Each node is a DEG/protein. Filled nodes represent the 3D structure of the protein. Empty nodes represent unknown 3D protein structures. The links between the proteins are the edges. Edges represent the protein-to-protein associations. Known interactions: from curated databases, experimentally determined. Predicted interactions: gene co-occurrence. Other interactions: textmining, co-expression, protein homology.

Figure 6. Protein-to-protein interaction networks for GSE9944 dataset. **A.** Top ten upregulated DEGs. **B**. Downregulated DEGs.



Each coloured circle represents a node. Each node is a DEG/protein. Filled nodes represent the 3D structure of the protein. Empty nodes represent unknown 3D protein structures. The links between the proteins are the edges. Edges represent the protein-to-protein associations. Known interactions: from curated databases, experimentally determined. Predicted interactions: gene co-occurrence. Other interactions: textmining, co-expression, protein homology.

Figure 7. Functional enrichment analysis bubble plots of the ten most differentially expressed genes. **A**. Upregulated DEGs for GSE27276 dataset. **B**. Downregulated DEGs for GSE27276 dataset. **C.** Upregulated DEGs for GSE9944 dataset.



BP: biological processes, CC: cellular compartments, MF: molecular function, KEGG: Kyoto Encyclopedia of Genes and Genomes pathway. Count represents the number of genes involved in each process.

Discussion

In the presented study, diverse bioinformatics tools were utilised to identify and analyse the ten most upregulated and downregulated DEGs from two glaucoma datasets obtained from the GEO webserver, corresponding to transcriptomic data from trabecular meshwork and astrocytes from the optic nerve head of glaucoma patients compared to a control group. The aim was to explore their possible role in the pathogenesis and progression of the disease and to identify potential molecular markers with diagnostic and therapeutic implications.

In the dataset GSE27276, corresponding to TM tissue, among all upregulated genes, four of them, HBA1, HBA2, HBB, and HBD, had the highest level of significance and formed a cluster during the STRING analysis. Its relation relies on that all encode different subunits of haemoglobin (Hb), an essential molecule in the oxygen and carbon dioxide transport. HbA is the most common type of Hb in adults and is composed of two alpha subunits encoded by HBA1 and HBA2, and two beta subunits encoded by HBB. HBD encodes the globin delta chain, which forms part of a small fraction of adult Hb, the Hb A2 type (Hoffbrand and Steensma, 2019), The overexpression of these genes may indicate a hypoxic state of the TM tissue. It is known that the TM of glaucoma patients present a major resistance to the aqueous humour drainage, which generates ocular hypertension, and an imbalance between the rate in its production and its elimination. The impairment in fluid renewal may contribute to the scarcity of nutrient supply, including oxygen. Moreover, ocular hypertension generates mechanical stress to the TM cells contributing to the generation of reactive oxidative species (ROS) (Liton and Gonzalez, 2008). Since Hb has antioxidant properties (Gabbianelli *et al.,* 1998), upregulation of its different subunits may also indicate the existence of a defensive mechanism against oxidative damage in addition to supporting nutrient supply.

However, one of the major problems in the TM during glaucoma is the loss of its physiological properties. CYTL1 participates in the differentiation of chondrocytes and MGP, both found upregulated, encodes a vitamin K-dependent protein involved in calcium inhibition. Its upregulation in the TM of individuals with POAG may suggest a protective mechanism to prevent calcification, thereby helping to preserve the flexibility of the TM and regulate its resistance. Moreover, Xue and colleagues (2006) demonstrated that in TM cells, MGP inhibits the bone morphogenetic protein 2 (BMP2), a secreted ligand of the transforming growth factor-beta (TGF-β), involved in bone formation. Furthermore, some studies have revealed high levels of TGF-β in the aqueous humour of POAG patients (Tamm and Fuchshofer, 2007) suggesting that the ECM might be undergoing a transformation process affecting its quality and quantity thus increasing

its resistance. In this scenario, MGP may play a key role as compensatory mechanism to counteract the effect of TGF-β and BMP2. Similarly, the upregulated FMOD, a gene that encodes the proteoglycan fibromodulin, and which main function is the assembly of ECM by binding to collagen type I, has also been implicated in the regulation of TGF-β activities by binding to it within the ECM (Zheng *et al*., 2023). Recent research has indicated the implication of FMOD in fibrotic processes, particularly under oxidative stress conditions, in animal models (An *et al*., 2020), and in patients with cardiac failure (Andenæs *et al*., 2018). This suggests that FMOD upregulation may be a direct causative agent of glaucoma progression, which might contribute to the closure or reduction of porosity in the TM by substituting health, soft TM tissue by a stiffness, fibrotic structure. Furthermore, PPI analysis has shown a direct relationship between FMOD and MGP. This might indicate that MGP is acting as a preventative mechanism against calcium deposit formation, contributing to the maintenance of the ECM integrity to preserve its physiological function, thereby mitigating the impact of FMOD and preventing fibrosis.

Closely related to immunity, HLA-DPA1 and FCGBP genes have been identified as upregulated in the TM. HLA-DPA1 is expressed only by antigen-presenting cells, and is part of the major histocompatibility complex class II, which mainly functions to present fragments of foreign antigens to naïve T cells. This gene has been linked to susceptibility of some chronic inflammatory diseases (Díaz-Peña *et al*., 2020) and associated with autoimmune conditions (Xie *et al*., 2013). Recent research has indicated that variants of this gene might be involved in angle-closure glaucoma (Li *et al.,* 2019), while other studies have suggested that glaucoma could potentially be an autoimmune disease, possibly mediated by T-cell autoimmunity (Wakefield and Wildner, 2020). FCGBP plays key roles as part of innate immunity by secreting mucin thus protecting mucosal surfaces from pathogens (Liu *et al*., 2022), including the ocular surface (Bräuer *et a*l., 2023). However, increased levels have been associated with intestinal inflammation (Liu *et al*., 2022) and with autoimmune diseases (Kobayashi *et al*., 2001). Although the exact mechanisms implicated in each are yet to be discovered and further research is needed to explore possible links between glaucoma and autoimmunity, if they exist.

The last significant overexpressed gene found in the TM is C2orF40, also known as ECRG4. This gene encodes a hormone-like protein, named augurin, which is involved in cellular senescence and acts as an inhibitor of cellular proliferation. It exhibits immunoregulatory functions by acting as an immune signalling molecule and is considered a sentinel protein (Qin and Zhang, 2019). The relation between this gene and glaucoma appears not to be elucidated

yet. However, Kujuro and colleagues (2010) demonstrated that upregulation of ECRG4 induced cell senescence in a mouse model of oligodendrocytes precursor cells due to a rapidly degradation of the cell cycle regulators, cyclins D1 and D3. Furthermore, Huh and collaborators (2009) found overexpression of ECRG4 during chondrogenesis in mice, while Ruggiero and coworkers (2022) identified increased expression in an *in vitro* model of osteoblast differentiation. The upregulation of this gene observed in the human TM during the execution of this analysis may be another indicative of the connective tissue transformation into a more cartilaginous structure. Further research is needed to validate this hypothesis and to examine ECRG4 levels in easily accessible fluids, such as tears. If these findings are confirmed, ECRG4 could become a valuable biomarker for the early detection of glaucoma.

In relation to the upregulated genes identified in the astrocytes of the OHN tissue, dataset GSE9944, ADH1A, ADH1B and RBP1 formed a cluster during the STRING examination, and GO analysis determined that these genes, along with AKR1C3, participate in the metabolism of retinol and, that all these DEGs except RBP1, are involved in retinol dehydrogenase activity. ADH1A, ADH1B are genes that encode the subunits alpha and beta, respectively, of the enzyme alcohol dehydrogenase. This enzyme transforms the alcohol into acetaldehyde, which without the further conversion into acetate by the aldehyde dehydrogenase (ALDH), acetaldehyde might accumulate and generate cellular damage by oxidative stress and ultimately, promote cellular death (Orywal and Szmitkowski, 2017). Since ALDH was not among the top ten upregulated genes, acetaldehyde may be contributing to the RGCs death. Additionally, ADH1A, ADH1B also participate in the conversion of retinol into retinal or retinaldehyde (Orywal and Szmitkowski, 2017). RBP1 encodes the protein responsible for the transport of retinol. Retinol is essential for the retina as is a key component of the rhodopsin, a pigment crucial for the low-light vision (Sajovic *et al*., 2022). Its upregulation could suggest a mechanism to ensure sufficient availability of retinol for these processes. However, given the increased expression of these genes, a potential raise in ROS levels may also be deteriorating the health of RGCs contributing into the progression of glaucoma. Similarly, the upregulation of DCLK1, identified in this study, may suggest that the ONH and its RGCs may be undergoing a repair process. DCLK1 is physiologically expressed in RGCs and plays a role in regulating axon growth during their maturation (Nawabi *et al*, 2015). AKR1C3 encodes an aldo-keto reductase, which participates in the metabolism of prostaglandins by catalysing the transformation of ketones and aldehydes into alcohols. Prostaglandins have a key role in the regulation of ocular tension as they facilitate the fluid outflow by inducing the expression of matrix metalloproteinases and relaxing the smooth muscle in the ciliary body. Metalloproteinases contribute to the remodulation of the ECM thereby

enhancing its porosity (Doucette and Walter, 2017). Overexpression of AKR1C3 may lead into a reduction of the prostaglandin levels due to their premature inactivation and as a result, it may cause an increase in the IOP due to an obstruction in the physiological ocular drainage system, contributing into the progression of glaucoma. In fact, prostaglandin analogues are currently the first therapeutic line in the management of the disease (Doucette and Walter, 2017). Given the role of AKR1C3 in prostaglandin metabolism, the regulation of this gene could be explored as a therapeutic target in glaucoma. Another therapy that has been explored to prevent ocular damage was the intake of selenium (Xu *et al*., 2023). Interestingly, SEPP1, a gene that produces the selenoprotein, the molecule in charge of transporting selenium in plasma, was also among the top ten upregulated genes in the astrocytes of the ONH. It is well known that selenium has antioxidant properties (Tinggi, 2008), and this characteristic has been linked to the prevention of diverse diseases including cataracts and macular degeneration (Khoo *et al*., 2019; Xu *et al*., 2023). The upregulation of SEPP1 in POAG patients compared to non-eye disease related controls might suggest that SEPP1 is acting as a defensive mechanism to protect the ONH against the oxidative stress to slow down the progression of disease.

Another upregulated gene that may be protecting the ONH is the KCTD12. This gene encodes an auxiliary protein that modulates the G-protein coupled receptor, more specifically, the gamma-aminobutyric acid type B receptor, a type of membrane protein that participates in cellular communication (Ivankova *et al*., 2013). KCTD12 regulates the excitation of neurons by promoting it during initial phases of neuronal maturation and inhibiting once maturation is completed (Chun *et al*., 2021). Its upregulation in glaucoma ONH astrocytes may indicate a response to stress, regulation of cellular homeostasis, or a neuroprotective mechanism exerted by the astrocytes. However, further investigation would be needed to understand the involvement of this gene in glaucoma and its molecular mechanisms. Further, MID1 also plays a key role during embryological development by encoding an E3-ubiquiting ligase. This ligase is associated with microtubules and is involved in the ubiquitination of target proteins (Baldini *et al*., 2020). Its overexpression in the ONH astrocytes in glaucoma patients may indicate multiple functions. On one hand, it may reflect an exaggerated need to degrade damaged proteins accumulated during the cellular stress. On the other hand, this upregulation might be aimed at regenerating the injured axons. Furthermore, Schweiger and colleagues (2017) identified overexpression of MID1 in other neurodegenerative diseases, such as Alzheimer´s disease. These researchers suggested that MID1 might have an essential role in the degradation of the protein phosphatase 2A. They proposed that hyperactivity of MID1 could lead to decreased

levels of the protein phosphatase 2A, resulting in a dephosphorylation impairment and the accumulation of hyperphosphorylated proteins, which may promote disease.

Immunity-related genes were also identified in the ONH, LY96 and CEBPD were among the top ten upregulated DEGs. LY96 encodes a protein that is expressed in cells of the innate immunity, such as macrophages. This molecule interacts with the Toll-like receptor 4 as a co-receptor to activate a downstream signalling leading to the production of pro-inflammatory mediators (Ferrari *et al*., 2024). This overexpression may result from the detection of damage-associated molecular patterns generated during the inflammatory cell death that occurs throughout the progression of glaucoma. CEBPD encodes a transcription factor, and it has been described as increased in the astrocytes of other neurodegenerative diseases, such as Alzheimer´s disease (Ko *et al*., 2015). Ko and colleagues (2015) discovered that CEBPD can reduce the process by which macrophages engulf damaged neurons. Considering this possibility in glaucoma, more cellular debris could be accumulated leading to increased inflammation, which is one of the key features of glaucomatous processes.

Regarding the top ten downregulated DEGs identified during this analysis, in the GSE27276 dataset, KRT19, CALML3, KRT15, KRT13, C19orf33, PAX6, SLPI, LCN2, S100A9 and S100A8 were identified as significant. These genes formed two clusters during the STRING analysis between SLPI, LCN2, S100A9 and S100A8 and between KRT19, KRT15 and KRT13. The GO analysis determined that the genes included in the latter cluster were associated with cellular structural organisation and epithelial differentiation, while the cluster formed by SLPI, LCN2, S100A9, and S100A8 was involved in immunity.

KRT13, KRT15 and KRT19 are genes that express intermediate filaments in epithelial cells and the downregulation detected in the TM of POAG patients compared to controls during this study suggests that the epithelial-like cells that form this specialised tissue are losing their physiological properties. Together with the upregulation of MGP and CYTL1, it appears that the TM is becoming less epithelial while gaining more cartilaginous consistency.

In relation to the downregulated genes involved in immune response, S100A8 and S100A9 express calcium-binding proteins that often form a complex, S100A8/A9, named calprotectin. These proteins are expressed by phagocytic cells, such as macrophages and neutrophils. They possess several functions within inflammatory and immune processes. For instance, have chemoattractant properties and promote the release of cytokines, but also are involved in the generation of ROS by favouring the activation of NADPH oxidase. Additionally, these genes

participate in the synthesis of arachidonic acid, which is essential to produce prostaglandins (Wang *et al*., 2018). Since prostaglandins play a key role in the TM homeostasis, the downregulation of S100A8 and S100A9 may contribute to the TM degeneration by hindering its production. On the contrary, this downregulation also minimises the generation of ROS thus reducing cellular stress and damage. This fact exposes the potential ambiguity of S100A8 and S100A9 downregulation that may act as an indirect protective mechanism, but on the other hand, contributing to the characteristic pathogenic process of glaucoma. More research would be required to investigate the exact involvement of these genes in glaucoma.

SLPI, directly linked to S100A9 and LCN2 during the PPI analysis, this gene was also one of the top ten downregulated expressed genes in the TM related to immunity. The GO analysis determined that SLPI is involved in the response to lipopolysaccharides and innate immunity. SLPI encodes a protease inhibitor, mainly secreted by epithelial cells, able to suppress the action of some enzymes expressed by cells of the innate immune system, such as neutrophils. By doing this, it protects the tissues from excessive damage caused by uncontrolled immune responses and excessive inflammation, while promoting tissue repair (Majchrzak-Gorecka *et al*., 2016). Its downregulation could indicate that TM cells are lacking this protection and are more vulnerable to the immune activity. Finally, the last gene involved in this cluster, LCN2, expresses a glycoprotein able to transport lipophilic molecules that participate in eye homeostasis such as retinoids and prostaglandins. It has a significant role in innate immunity and modulating inflammation, which confers a protective action from cellular stress (Jaberi *et al*., 2021). The downregulation of this gene in the TM might indicate that this structure is being more vulnerable to the detrimental action of inflammatory and oxidative species contributing to the progression of the disease. In addition, LCN2 can bind to the matrix metalloproteinase 9 (MMP-9) forming a complex that controls the action of MMP-9 and prevents this molecule from degrading the ECM (Matet *et al.,* 2020). The reduction in the expression of LCN2 might create a disequilibrated environment leading to an excessive ECM destruction and consequently, interfering in the eye fluid outflow through the TM pathway. Interestingly, Matet and colleagues (2020) observed an unexpected reduction in serum levels of LCN2 and the LCN2/MMP9 complex in patients with chorioretinopathy and they manifested that, as of 2020, it was the sole eye condition linked to a reduced level of LCN2 expression. They also pointed out that LCN2 serum levels could have utility as a biomarker for that disease. However, it would be interesting to evaluate serum levels of LCN2 also in glaucoma patients to explore potential similarities and gain a better understanding of this gene in the context of eye pathology.

Interestingly, the PAX6 gene encodes an essential transcription factor well-known for its key role in the development of eye structures during foetal growth and it maintains the stem cell reservoir for the regeneration of ocular tissues (Manuel *et al*., 2015). The downregulation of this gene in TM is the significant importance as indicates a compromised ability for tissue regeneration and repair. Moreover, MID1 controls the tagging of PAX6 for degradation by the proteosome through its ubiquitination (Pfirrmann *et al*., 2016). MID1 was identified as upregulated in the astrocytes of the ONH during this experiment and this finding might suggest that the overexpression may interfere with the expression of PAX6, thereby reducing its levels in the TM. However, the detected upregulation of MID1 and downregulation of PAX6 were observed in two different cohorts. To evaluate the relationship of these two proteins, it would be necessary to examine their expression in both tissues from the same cohort. Similarly, C19orf33, also known as immortalization-upregulated protein, whose main function is to interfere with the cell cycle to promote cell proliferation (Luo *et al.*, 2023), has been identified as one of the top ten downregulated genes in TM tissue during this analysis. This downregulation in glaucoma may evidence a reduction in the ability of the TM cells to proliferate, which could contribute to impaired tissue repair and regeneration associated with the disease.

A gene that could provide some therapeutic relief by modulating its activity is the CALML3 gene, found as downregulated in this experiment. This gene encodes a protein resident in the extracellular exosome of epithelial cells and plays a key role in calcium homeostasis. Pu and colleagues (2018) demonstrated that overexpression of CALML3 in epithelial cells of the lens reduced apoptosis in an *in vitro* model of oxidative stress induced by hydrogen peroxide. Calcium dysregulation and an excessive intracellular calcium accumulation are known to lead to oxidative stress, exacerbating the ROS production, which in turn compromise the mitochondrial activity, and ultimately, promotes apoptosis through the activation of the caspase cascade (Crish *et al.,* 2011). These researchers concluded that elevated levels of CALML3 offer a protective mechanism against programmed cell death and hypothesised that this may result from the role of CALML3 in controlling the generation of ROS by preventing excessive calcium accumulation inside the cells, thereby inhibiting the activation of the caspase pathway. Following this hypothesis, the low levels of CALML3 seen during this study may manifest a cellular vulnerability in POAG patients to oxidative stress. Artificially increasing the activity of this calcium modulator may provide beneficial effects by slowing down the progression of the disease.

In the case of the ONH, three significant downregulated genes were detected in the astrocytes, ENO2, ELN, and BMP1. These last two showed mutual interaction during the STRING analysis.

ELN encodes a structural protein, elastin, which forms part of the ECM and is responsible for the elasticity and flexibility of soft tissues (Trębacz and Barzycka, 2023). Miller and coworkers (2009) observed a notable reduction in the elasticity of cortical astrocytes following mechanical trauma *in vitro* and posteriorly Hariani and colleagues (2024) showed in a rat model a reduced level of elastin in reactive astrocytes from ONH, which also corresponded with a low level of the enzyme that mediates the elastin synthesis, lysyl oxidase like 1 (LOXL1). These discoveries along with the downregulation finding of ELN during this study suggests that the same might be occurring during human glaucomatous processes. However, more evidence, including laboratory experimentation, would be needed to demonstrate this fact. Additionally, BMP1, which was also significantly downregulated and linked to ELN by the STRING analysis, encodes a proteolytic enzyme, a metalloprotease, that is involved in the transformation of elastin precursors into their mature form (Rosell-García *et al*., 2022). Interestingly, in the pseudoexfoliation syndrome, a well-known cause of one of the open-angle glaucoma subtypes, pseudoexfoliation glaucoma, ELN, in addition to LOXL1 and other elastic fibre-related compounds, were found downregulated in the lamina cribosa area. Furthermore, elastin and LOXL1 among other compounds derived from elastic fibres were identified interfering the eye fluid drainage pathway (Rosell-García *et al*., 2022). Considering all these facts together, it is logical to think that there is a relationship between the downregulation of ELN and other compounds involved in the synthesis of elastic fibres, such as BMP1, in the genesis and progression not only in pseudoexfoliation glaucoma subtype, but also in POAG and potentially other types.

ENO2 encodes an enzyme, enolase, that participates actively in the glycolysis pathway. It is also referred to as neuron-specific enolase and it provides protection to the neuronal and glial cells (Haque *et al*., 2018). ENO2 is used as a serum biomarker for neuronal damage during ischemic brain injury, stroke and traumatic brain damage among others where it is found upregulated (Jiang *et al*., 2021). However, Jiang and colleagues (2021) demonstrated a focal downregulation of ENO2 three hours following cerebral ischemia in a rodent model. The low levels of ENO2 found in the cultured astrocytes of glaucoma patients during this study may, therefore, indicate a constant and continuous injury in these cells which would reflect a compromised metabolism and ultimately an impaired function. Furthermore, Dai *et al.* (2022) utilised a model of machine learning to identify that the downregulation of ENO2 could serve as a diagnostic biomarker for glaucoma and might also be a viable pharmacological target.

Lastly, it is crucial to acknowledge the limitations of the presented study. The small sample size and limited number of donors may constrain the results. Access to a larger a more diverse

sample and donor size would enhance the accuracy and reliability of the findings. Additionally, the lack of demographic information such as therapeutic treatments and comorbidities at the time of sample collection that could interfere in the expression of genes and thus impact the results. Finally, the lack of experimental data, as this study was solely based on free information published in recognised scientific databases and access-free webservers to perform the bioinformatic analysis.

Conclusion

This study identified significant DEGs with involvement in glaucoma in both the TM and in the astrocytes of the ONH and provided insights into their potential role in the disease. Overall, DEGs detected were related to haemoglobin (HBA1, HBA2, HBB, and HBD), to structural and extracellular matrix processes (MGP, FMOD, KRT19, KRT15, KRT13, ELN, and BMP1), immunity-related (HLA-DPA1, FCGBP, SLPI, LCN2, and LY96), associated to neural and developmental functions (CYTL1, PAX6, C19orf33, DCLK1, KCTD12, and MID1), involved in calcium signalling (CALML3), in transcription regulation (CEBPD), and associated with metabolic processes (AKR1C3, SEPP1, ADH1B, ADH1A, RBP1, and ENO2). Additionally, there were DEGs with hybrid functions involving immune responses, cellular proliferation, and metabolic processes (C2orf40 S100A9, and S100A8). While any of these DEGs could be further explored as potential therapeutic target, priority should be given to AKR1C3, CALML3 and ENO2 due to their roles and the implications observed in this study. Additionally, ENO2 could also be investigated as potential diagnostic biomarker. However, limitations must be considered, and further research and experimental studies are required to evaluate the proposed DEGs and elucidate their implications in glaucoma.

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Appendix

Table A1. GSE27276 demographic data of donors (Liu *et al.,* 2013).

Group	Accession	Title	Sex	Age (Years)	Ethnic	Tissue	Post-Morten Delay (hours)
Patient	GSM674408	POAG 1	Unspecified	Unspecified	Unspecified	TM	0
	GSM674409	POAG 2	Unspecified	Unspecified	Unspecified	TM	0
	GSM674410	POAG 3	Unspecified	Unspecified	Unspecified	TM	0
	GSM674411	POAG 4	Unspecified	Unspecified	Unspecified	TM	0
	GSM674431	POAG 5	Unspecified	Unspecified	Unspecified	TM	0
	GSM674432	POAG 6	Unspecified	Unspecified	Unspecified	TM	0
	GSM674433	POAG 7	Unspecified	Unspecified	Unspecified	TM	0
	GSM674434	POAG 8	Unspecified	Unspecified	Unspecified	TM	0
	GSM674435	POAG 9	Unspecified	Unspecified	Unspecified	TM	0
	GSM674436	POAG 10	Unspecified	Unspecified	Unspecified	TM	0
	GSM674437	POAG 11	Unspecified	Unspecified	Unspecified	TM	0
	GSM674438	POAG 12	Unspecified	Unspecified	Unspecified	TM	0
	GSM674439	POAG 13	Unspecified	Unspecified	Unspecified	TM	0
	GSM674440	POAG 14	Unspecified	Unspecified	Unspecified	TM	0
	GSM674441	POAG 15	Unspecified	Unspecified	Unspecified	TM	0
	GSM674442	POAG 16	Unspecified	Unspecified	Unspecified	TM	0
	GSM674443	POAG 17	Unspecified	Unspecified	Unspecified	TM	0
Control	GSM674412	Control 1	Unspecified	Unspecified	Unspecified	TM	5.24
	GSM674413	Control 2	Unspecified	Unspecified	Unspecified	TM	5.42
	GSM674414	Control 3	Unspecified	Unspecified	Unspecified	TM	5.24
	GSM674415	Control 4	Unspecified	Unspecified	Unspecified	TM	4.18
	GSM674416	Control 5	Unspecified	Unspecified	Unspecified	TM	5.04
	GSM674417	Control 6	Unspecified	Unspecified	Unspecified	TM	4.5
	GSM674418	Control 7	Unspecified	Unspecified	Unspecified	TM	6.24
	GSM674419	Control 8	Unspecified	Unspecified	Unspecified	TM	5.42
	GSM674420	Control 9	Unspecified	Unspecified	Unspecified	TM	4.07
	GSM674421	Control 10	Unspecified	Unspecified	Unspecified	TM	5.38
	GSM674422	Control 11	Unspecified	Unspecified	Unspecified	TM	3.24
	GSM674423	Control 12	Unspecified	Unspecified	Unspecified	TM	3.43
	GSM674424	Control 13	Unspecified	Unspecified	Unspecified	TM	4.53
	GSM674425	Control 14	Unspecified	Unspecified	Unspecified	TM	3.17
	GSM674426	Control 15	Unspecified	Unspecified	Unspecified	TM	4.18
	GSM674427	Control 16	Unspecified	Unspecified	Unspecified	TM	5.04
	GSM674428	Control 17	Unspecified	Unspecified	Unspecified	TM	4.5
	GSM674429	Control 18	Unspecified	Unspecified	Unspecified	TM	6.24
	GSM674430	Control 19	Unspecified	Unspecified	Unspecified	TM	2.25

Table A2. GSE9944 demographic data of donors (Lukas *et al.,* 2008),

Group	Accession	Title	Sex	Age (Years)	Ethnic	Disease State	Tissue
Patient	GSM251599	00-6R-rep1	F	56	C	Glaucoma 1	ONH
	GSM251600	00-7L-rep1	M	83	C	Glaucoma 2	ONH
	GSM251601	00-7R-rep1	M	83	C	Glaucoma 3	ONH
	GSM251602	02-1R-rep1	M	69	C	Glaucoma 4	ONH
	GSM251603	2-10R-rep1	F	80	C	Glaucoma 5	ONH
	GSM251604	04-1L-rep1	F	79	C	Glaucoma 6	ONH
	GSM251605	04-5A-rep1	F	72	C	Glaucoma 7	ONH
	GSM251613	04-20L-rep1	M	67	AA	Glaucoma 8	ONH
	GSM251614	04-20L-rep2	M	67	AA	Glaucoma 9	ONH
	GSM251615	04-24R-rep1	F	53	AA	Glaucoma 10	ONH
	GSM251616	04-24R-rep2	F	53	AA	Glaucoma 11	ONH
	GSM251617	05-16L-rep1	F	56	AA	Glaucoma 12	ONH
	GSM251618	05-16L-rep2	F	56	AA	Glaucoma 13	ONH
Control	GSM251607	99-9R-rep1	F	46	C	Control 1	ONH
	GSM251608	LC46R-rep1	M	46	C	Control 2	ONH
	GSM251609	99-6L-rep1	M	42	C	Control 3	ONH
	GSM251610	03-11R-rep1	F	68	C	Control 4	ONH
	GSM251611	03-13L-rep1	F	72	C	Control 5	ONH
	GSM251612	04-2L-rep1	M	65	C	Control 6	ONH

Highlighted donors from which samples were taken from both eyes. Each colour represents one individual.

C, Caucasian; AA, African American.

Table A3. Functional Annotation Analysis Raw Data

		Term Code	Term Description	Genes Involved	P-value	Fold Enrichment	FDR
GSE27276	Upregulated	GO:0042542	Response to hydrogen peroxide	HBB, HBA2, HBA1	1.11E-04	164.58	6.48E-04
		GO:0015670	Carbon dioxide transport	HBB, HBA2, HBD, HBA1	1.22E-11	764.13	3.89E-10
		GO:0015671	Oxygen transport	HBB, HBA2, HBD, HBA1	2.22E-11	668.61	3.89E-10
		GO:0042744	Hydrogen peroxide catabolic process	HBB, HBA2, HBD, HBA1	2.49E-10	382.06	2.91E-09
		GO:0098869	Cellular oxidant detoxification	HBB, HBA2, HBD, HBA1	1.55E-08	140.76	1.36E-07
		GO:0030185	Nitric oxide transport	HBB, HBA2, HBA1	1.51E-06	1283.73	1.06E-05
		GO:0031838	Haptoglobin-haemoglobin complex	HBB, HBA2, HBD, HBA1	1.22E-11	789.27	1.96E-10
		GO:0072562	Blood microparticle	HBB, HBA2, HBD, HBA1	3.10E-07	69.80	2.37E-06
		GO:0005833	Haemoglobin complex	HBB, HBA2, HBD, HBA1	1.70E-11	732.89	1.96E-10
		GO:0071682	Endocytic vesicle lumen	HBB, HBA2, HBA1	3.24E-05	307.82	1.86E-04
		GO:0070062	Extracellular exosome	FCGBP, MGP, HBB, HBA2, HBA1	0.011	4.58	0.037
		GO:0005615	Extracellular space	FCGBP, HBB, HBA2, CYTL1, HBA1, FMOD	6.69E-04	6.38	0.003
		GO:0031012	Extracellular matrix	FCGBP, MGP, FMOD	0.004	28.50	0.014
		GO:0005344	Oxygen transporter activity	HBB, HBA2, HBD, HBA1	4.32E-11	590.09	4.10E-10
		GO:0019825	Oxygen binding	HBB, HBA2, HBD, HBA1	1.56E-09	255.18	9.88E-09
		GO:0031720	Haptoglobin binding	HBB, HBA2, HBD, HBA1	8.97E-09	755.32	4.26E-08
		GO:0004601	Peroxidase activity	HBB, HBA2, HBD, HBA1	3.02E-07	251.77	1.15E-06
		GO:0020037	Haem binding	HBB, HBA2, HBD, HBA1	5.33E-07	60.91	1.69E-06
		GO:0043177	Organic acid binding	HBB, HBA2, HBD, HBA1	7.84E-12	858.32	1.49E-10
		GO:0031721	Haemoglobin alpha binding	HBB, HBD	3.03E-06	944.15	8.21E-06
		hsa05144	Malaria	HBB, HBA2, HBA1	1.94E-04	103.94	0.003
		hsa05143	African trypanosomiasis	HBB, HBA2, HBA1	1.06E-04	140.46	0.003
GSE9944		GO:0042572	Retinol metabolic process	ADH1B, ADH1A, RBP1, AKR1C3	1.16E-06	158.49	1.03E-04
		GO:0004745	Retinol dehydrogenase activity	ADH1B, ADH1A, AKR1C3	2.97E-05	314.72	0.002
		hsa04936	Alcoholic liver disease	ADH1B, ADH1A, LY96	0.002	36.60	0.025

Biological Processes | Cellular Component | Molecular Function | KEGG pathway

Table A3. Functional Annotation Analysis Raw Data (continuation).

		Term Code	Term Description	Genes Involved	P-value	Fold Enrichment	FDR
GSE27276	Downregulated	GO:0045109	Intermediate filament organization	KRT19, KRT15, KRT13	2.63E-04	104.65	0.016
		GO:0030855	Epithelial cell differentiation	KRT19, KRT15, KRT13	5.30E-04	73.68	0.016
		GO:0045087	Innate immune response	SLPI, LCN2, S100A9, S100A8	6.61E-04	18.23	0.016
		GO:0032496	Response to lipopolysaccharide	SLPI, S100A9, S100A8	0.001	51.95	0.017
		GO:0042742	Defence response to bacterium	LCN2, S100A9, S100A8	0.002	43.24	0.018
		GO:0005856	Cytoskeleton	KRT19, KRT15, KRT13, S100A9, S100A8	4.85E-05	19.51	6.31E-04
		GO:0005882	Intermediate filament	KRT19, KRT15, KRT13	0.002	43.97	0.010
		GO:0070062	Extracellular exosome	KRT19, SLPI, KRT15, KRT13, LCN2, CALML3, S100A9, S100A8	5.41E-06	7.33	1.41E-04
		GO:0005198	Structural molecule activity	KRT19, KRT15, KRT13	0.003	32.61	0.038
		hsa04915	Oestrogen signalling pathway	LCN2, S100A9, S100A8	1.29E-04	31.61	0.006
		hsa05150	Staphylococcus aureus infection	KRT19, KRT15, KRT13	0.002	33.84	0.039
		hsa04657	IL-17 signalling pathway	KRT19, KRT15, KRT13, CALML3	0.002	34.56	0.039

Biological Processes Cellular Component Molecular Function KEGG pathway